MPPT MODBUS ADDON

Workflow Overview

End-to-End Data Pipeline Reference

WC-MA-WO-v1.0 | 2026-03-09

Table of Contents

1. Purpose

This document provides a concise end-to-end overview of how the WakeCap MODBUS Addon Board communicates with an EPEver MPPT solar charge controller, collects register data, assembles it into a compact TLV packet, and delivers it over the Wirepas mesh network.

It is intended for firmware and backend engineers who need to understand the data pipeline without reading source code.

2. Modbus Basics

Modbus is an industrial serial protocol originally published by Modicon in 1979. The WakeCap addon uses Modbus RTU (binary-encoded) over an RS-485 physical layer.

2.1 Key Concepts

Concept	Description
Master / Slave	Addon board is the master (initiates). MPPT controller is the slave (responds).
Device ID	Each slave has a unique address. EPEver uses Device ID = 0x01.
Function Code 0x04	Read Input Registers — real-time measurements and status.
Function Code 0x03	Read Holding Registers — configuration data (e.g., RTC).
Register	16-bit (2-byte) data unit addressed by a 16-bit number.
CRC-16	Every frame ends with a 2-byte CRC for error detection (poly 0xA001).

2.2 Frame Format

Request (Master → Slave), 8 bytes:

```
[DevID] [FuncCode] [AddrHi] [AddrLo] [CountHi] [CountLo] [CRC_Lo] [CRC_Hi]
```

Response (Slave → Master), variable length:

```
[DevID] [FuncCode] [ByteCount] [Data ...] [CRC_Lo] [CRC_Hi]
```

2.3 Physical Interface

Parameter	Value
Standard	RS-485 (half-duplex, differential pair)
Baud Rate	115,200 bps
Data Format	8N1 (8 data bits, no parity, 1 stop bit)
Transceiver	MAX485 (or equivalent)
TX Pin	GPIO 15
RX Pin	GPIO 14
RE Pin	GPIO 16 (Receiver Enable — active low)

3. System Architecture

The addon board sits between the MPPT controller (RS-485 side) and the Wirepas mesh (radio side). It periodically polls the MPPT, packages the data, and transmits it wirelessly.

```
┌──────────────┐    RS-485     ┌──────────────────┐
│   EPEver     │◄────────────► │ WakeCap MODBUS   │
│   MPPT       │    2-wire     │ Addon Board      │
│  Controller  │  half-duplex  │ (nRF52832)       │
└──────────────┘               └──────────────────┘
                                        │  Wirepas Mesh
                                        │  (2.4 GHz)
                                        ▼
                               ┌──────────────────┐
                               │ Wirepas Mesh     │
                               │ Network          │
                               └──────────────────┘
                                        │
                                        ▼
                               ┌─────────────────┐
                               │ Sink Node       │──► Gateway ──► MQTT
                               └─────────────────┘
```

4. Startup Sequence

1. Hardware init — UART and GPIO pins configured, RS-485 transceiver enabled.

2. Auto-detection — Board probes the bus to detect MPPT or Weather Station:

- Sends test read to register 0x311A (SOC) with function code 0x04.
- Response → MPPT detected. No response → probes 0x0000 with FC 0x03 for Weather Station.
- Timeout per probe: 500 ms, up to 3 retries per device type.

3. Scheduler start — Periodic polling task registered: 10 s initial delay, then every 60 seconds.

5. Register Polling Sequence

Each poll cycle reads 7 register groups sequentially from the MPPT controller:

#	Group	Address	FC	Regs	Bytes	Description
R0	Status	0x3200	0x04	3	6	Battery / Charging / Discharging status
R1	SOC	0x311A	0x04	1	2	Battery state of charge (%)
R2	Battery V/I	0x331A	0x04	2	4	Battery voltage and current
R3	Temps	0x3110	0x04	2	4	Battery temp and device temp
R4	PV V/I	0x3100	0x04	2	4	PV array voltage and current
R5	Load V/I	0x310C	0x04	2	4	Load output voltage and current
R6	RTC	0x9013	0x03	1	2	Real-time clock (minutes:seconds)

5.1 Timing Per Register

Parameter	Value
TX frame duration	~0.7 ms (8 bytes at 115,200 baud)
Response timeout	150 ms per register
Inter-register gap	100 ms
Time per register	~250 ms (typical)
Total poll time (7 registers)	~1.75 seconds worst-case

5.2 RS-485 Direction Switching

1. Assert RE high — enables transmitter, disables receiver.

2. Send 8-byte request frame (IRQ disabled during TX for clean timing).

3. Wait ~120 µs — transceiver settling time.

4. Assert RE low — enables receiver, disables transmitter.

5. Receive response bytes. Inter-frame gap (~2 ms idle) = end of frame.

6. Validate CRC against last 2 bytes of received data.

6. TLV Packet Construction

After all 7 registers are polled, their data is assembled into a single 29-byte TLV (Tag-Length-Value) packet:

6.1 Packet Layout

Byte	Field	Size	Value
0	TAG	1 B	0x01
1	LENGTH	1 B	0x1B (27)
2	BITMASK	1 B	Bit N = register N success
3–8	R0 Status	6 B	Battery + Charging + Discharging status
9–10	R1 SOC	2 B	State of charge (%)
11–14	R2 Batt V/I	4 B	Battery voltage + current
15–18	R3 Temps	4 B	Battery temp + device temp
19–22	R4 PV V/I	4 B	PV voltage + current
23–26	R5 Load V/I	4 B	Load voltage + current
27–28	R6 RTC	2 B	Minutes (high byte) + Seconds (low byte)

6.2 Bitmask

```
Bit:  7     6     5     4     3     2     1     0
      Rsvd  R6    R5    R4    R3    R2    R1    R0
            RTC   Load  PV    Temp  Batt  SOC   Status
```

> **ℹ NOTE**
> Bit = 1 means register data is valid. Bit = 0 means the read failed and the data bytes are zero-filled. Always check the bitmask before interpreting values.

6.3 Data Encoding

All 16-bit values are stored big-endian (MSB first). Scaling:

Field	Bytes	Scaling	Unit
Battery Status	3–4	Bitfield	—
Charging Status	5–6	Bitfield	—
Discharging Status	7–8	Bitfield	—
SOC	9–10	Direct (×1)	%
Battery Voltage	11–12	Raw × 0.01	V
Battery Current	13–14	Raw × 0.01	A
Battery Temp	15–16	Raw × 0.01	°C
Device Temp	17–18	Raw × 0.01	°C
PV Voltage	19–20	Raw × 0.01	V
PV Current	21–22	Raw × 0.01	A
Load Voltage	23–24	Raw × 0.01	V

Load Current	25–26	Raw × 0.01	A
RTC Minutes	27	Direct	min
RTC Seconds	28	Direct	sec

For any query, contact Zishan Shahzad, zishan.shahzad@wakecap.com

7. Mesh Delivery

7.1 Wirepas Endpoints

Endpoint	Size	Content
EP 61 (Data)	29 bytes	TLV packet — Tag + Length + Bitmask + 26 data bytes
EP 65 (Error)	3 bytes	LED state + timeout count + CRC error count

7.2 Error Status Packet (EP 65)

Byte	Field	Range	Meaning
0	LED State	0–2	0 = Normal, 1 = Problem, 2 = Critical
1	Timeout Count	0–255	Register timeouts since last report (resets after send)
2	CRC Error Count	0–255	CRC failures since last report (resets after send)

7.3 Backend Path

```
Addon Board → Wirepas Mesh → Sink Node → UART/SLIP → Gateway → MQTT → Backend
```

MQTT Topic: `<env>/gw-event/received_data/<gw-id>/<sink-id>/<net_id>/61/61`

8. Error Handling

Error Type	Detection	Recovery
No response (timeout)	150 ms timer expires	Zero-fill register, clear bitmask bit, increment counter
CRC mismatch	Calculated ≠ received CRC	Discard response, zero-fill, increment CRC counter
UART framing error	Hardware flags (overrun, break)	Treated as timeout
Device not found	Auto-detect fails (3 retries)	Default to MPPT, LED = Critical

9. Complete Poll Cycle Timeline

Phase	Duration
Boot + auto-detect	~3 seconds
Initial delay before first poll	10 seconds
Poll all 7 registers	~1.75 seconds
TLV assembly	< 1 ms
Wirepas mesh transmit (queuing)	< 1 ms
LED blink indication	~300 ms
Idle until next poll	~58 seconds
Total cycle interval	60 seconds

10. Quick Reference

Constant	Value
TLV Tag	0x01
TLV Length	0x1B (27 bytes)
Total Packet Size	29 bytes
Data Endpoint	EP 61
Error Endpoint	EP 65
Poll Interval	60 seconds
Modbus Device ID	0x01
Baud Rate	115,200 bps

Related Documents

Document	ID
MPPT Data Parsing Contract	WC-MA-DPC-v1.0
EPEver Solar Controller Protocol	Protocol v2.5 (manufacturer)
Wirepas Mesh SDK Documentation	Wirepas SDK

For any query, contact Zishan Shahzad, zishan.shahzad@wakecap.com